Community Hiding by Link Perturbation in Social Networks

Xianyu Chen, Zhongyuan Jiang, Hui Li, Jianfeng Ma, and Philip S. Yu, *Fellow, IEEE*

***Abstract*— Complex social network is a kind of relationship system composed of many nodes according to social relations. Community detection helps scholars to understand this network topology and find out meaningful communities. Many scholars are therefore actively exploring new community detection algorithms. However, it brings privacy issues such as the disclosure of personal or group information of community members and goes against an individual or group desire to be hidden. Hence, how to hide a target community in a network to resist the community detection algorithms becomes critical. Given this, this article studies the community hiding problem, which has not been extensively focused so far and aims to properly hide a target community into other communities by perturbing a budget limited number of social links. First, we formalize the community hiding problem and design a gain function. Second, we prove the feasibility of link perturbation operations and propose an efficient algorithm to solve the community hiding problem. Finally, we conduct extensive experiments on varieties of real small- and large-scale social networks and compare our method with other community deception algorithms. The experimental results demonstrate that the proposed algorithm is more efficient than previously proposed algorithms.**

***Index Terms*— Community deception, community detection, community hiding, community security, link perturbation.**

I. Introduction

COMPLEX networks are abstracted from a large number of complex systems that exist in reality. The overall function of a network is reflected in the interaction between each pair of nodes in the network. The community structure is its key structural feature and the communities correspond to the functional modules of the system. Extracting these modules of the network helps us deeply explore the internal building rules of complex networks. Therefore, researchers are more inclined to study how to efficiently discover the communities' structure of complex networks and pay a lot of attention to discover potential clustering information in social networks at the cost of the disclosure of sensitive community information. With the rise of awareness of personal privacy protection, we now focus on the protection of community privacy in social networks rather than exploring the potential community information in the networks. In real life, as each of us cannot exist alone without other people or other things, the result is that we are in many social networks with specific meanings, such as the patient network, mail network, cooperation network, and social network. Take a classic example to illustrate the importance of community hiding: police officers need to collect crime evidence in a social network as undercover police officers, and then, they must meet the following two requirements. First, the undercover police officers need to conduct normal social activities as normal social users, so as not to cause criminals to perceive them and destroy evidence or escape. Second, the communication among the police officers is usually frequent and it will form a police-community structure in social networks. However, this kind of community structure is not conducive to hiding the identity of the undercover police officers because of the high vigilance of criminals. Therefore, given the nature of the undercover police officers, we must find a way to destroy the community structure and successfully hide the police-community structure. Thus, it can be seen that how to hide one or more *target communities* in a network to evade social network analysis tools is extremely significant. This article aims to solve the single *target community* privacy disclosure issue and propose an efficient community hiding method.

What are the ideal goals for the hidden results of a community hiding method? Reference [1] gives three detailed ideal goals to measure community security. Given a social network G and a *target community* C, good hiding of community C in G has the following three manifestations.

1) *Reachability Preservation:* The members of C should be in the same connected component to reachable from one another to keep the information exchange, that is, link perturbation updates cannot break the *target community* C connectivity.
2) *Community Spread:* The members of C are supposed to be spread over other communities in G as much as possible.
3) *Community Hiding:* The members of C should be distributed in the largest communities.

These goals are also the guiding ideology for our work: on the basis of not destroying the connected component of the *target community*, in this work, we design a more efficient link perturbation mechanism to hide a specific *target community* into as many as possible other communities of a network.

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Xianyu Chen, Zhongyuan Jiang, Hui Li, and Jianfeng Ma are with the School of Cyber Engineering, Xidian University, Xi'an 710071, China (e-mail: xychen_3@stu.xidian.edu.cn; zyjiang@xidian.edu.cn; hli@xidian.edu.cn; jfma@mail.xidian.edu.cn).

Philip S. Yu is with the Department of Computer Science, University of Illinois at Chicago, Chicago, IL 60607 USA (e-mail: psyu@uic.edu).

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This article is dedicated to solve the community hiding problem, which has not been extensively focused so far and propose a new and efficient algorithm to achieve the three goals above. We focus on hiding a *target community C* into other communities by perturbing a budget limited number of social links. To perturb the most suitable links, we formalize the community hiding problem and design a gain function to guide perturbation operations. Also, we prove the feasibility and operability of the method. Finally, we prove the effectiveness of our community hiding algorithm through extensive experiments.

To solve the community hiding problem, our work makes the following contributions.

A. Propose an Approach to Evaluating the Strength of Dispersion Within a Community

An optional and effective way to prevent a node set from being detected into a community is to weaken the connections between nodes in the set. In this way, how to evaluate the strength of connections between each pair of nodes in such a node set? Here, we introduce an evaluation scheme to evaluate the strength of dispersion and the closeness of a community. In terms of community hiding algorithm, what we need to consider is the degree of dispersion of a community rather than the closeness of a community. Therefore, we introduce a *community dispersion* concept to evaluate the strength of dispersion within a community which is introduced in Sections III and IV.

B. Devise an Instantiation of the General Gain Function

To demonstrate the different effects of different link perturbation updates and guide the community hiding algorithm to carry out the most appropriate link perturbation operation, we formulate the community hiding problem and devise a new instantiation of the gain function. The detailed design of this gain function will be described in the subsequent sections.

C. Eliminate the Negative Effects of [1] in Two Particular Cases

Fionda and Pirro [1] proposed a community deception algorithm *Ds* and it treats link additions and deletions separately. Link additions include $inter\text{-}C$ (intercommunity) and $intra\text{-}C$ (intracommunity) link additions, and link deletions include $inter\text{-}C$ and $intra\text{-}C$ link deletions. Although the authors prove the feasibility of these four conditions on community deception, they cannot avoid the negative effects in case $intra\text{-}C$ link additions and case $inter\text{-}C$ link deletions on community deception. To solve the problem, we propose a more effective solution. First, we improve the *Ds* algorithm and form a new gain formula. Second, we prove the hidden impact of the above four conditions on community hiding and successfully prove the feasibility of operation $intra\text{-}C$ link additions and $inter\text{-}C$ link deletions. Hence, the most important and greatest contribution of this article is that our method completely excludes the adverse effects of the above two particular cases.

D. Extensive Experiments

To compare our method with previous methods and show the effectiveness of our community hiding methods, we conduct a lot of comprehensive experiments in a set of real social networks, including both small- and large-scale networks. For different social networks with different sizes, we will study in detail the influence of different community hiding methods on the community structure.

The rest of this article is organized as follows. Section II reviews the related research on community detection algorithms and community deception algorithms on complex social networks. Section III shows the preliminaries and problem formulation. We demonstrate the analysis and community hiding algorithm in Section IV. Section V gives the experimental evaluation. Finally, we summarize this article and describe the potential future work in Section VI.

II. Related Work

A. Community Detection Algorithms

Community detection algorithms are mature and a large number of community detection algorithms have been proposed since the community detection problem brought up. It is mainly divided into the following two categories: nonoverlapping and overlapping community detection. Here, we mainly give a brief introduction of the above two strategies separately in the following.

First of all, nonoverlapping community structure means that each node in network can belong to only one community, and there is no intersection between the community and the community. The classical algorithms of nonoverlapping detection methods include modularity optimization algorithms [2]–[4], affinity propagation algorithms [5], spectrum analysis algorithms [6], [7], random walk algorithms [8], [9], label propagation algorithms [10]–[12], and seed spread algorithms [13]–[15]. Among them, the modularity optimization algorithm is popular, which is first presented by Newman and Girvan [3], and modularity is a measure of the quality of the community division.

Overlapping community structures are closer to real-world network topologies. There are also some mature overlapping community detection methods. Palla *et al*. studied a cluster percolation method that is low efficiency in spare networks. Zhang *et al*. [17] made a detection survey based on multilabel propagation. Besides, the neighborhood-inflated seed expansion algorithm (NISE) is proposed by Whang *et al*. [18] to detect the overlapping communities. Furthermore, Ronhovde and Nussinov [19] proposed a community detection method that is based on the minimization of the Hamiltonian of Potts model and so on.

B. Community Hiding and Community Antidetection Algorithms

Due to the gradual improvement of people's awareness of privacy protection, some researchers started focusing on the community security problems in recent years. Algorithms include k-structural diversity algorithm [20]; k-degree

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TABLE I
COMPARISON WITH RELATED WORK

Paper	Operation	Method	Network Knowledge
Nagaraja[25]	+ links	Modularity	Vertex centrality
Waniek et al.[26]	+ or - links	Modularity	Community C's links
Fionda & Pirro[1]	+ or - links	Safeness1 & Modularity	Community C's links
This paper	+ or - links	Safeness2	Community C's links

anonymity algorithm [21], k-isomorphism algorithm [22], and Walk2Privacy algorithm [23]. Besides, Chen *et al.* [24] proposed a genetic algorithm (GA) to search effective perturbation attack schemes for leading privacy protection. In terms of community hiding, Nagaraja [25] first proposed the idea of hiding a community to resist community detection algorithm and the studies have shown that perturbing some links can effectively hide a community. However, Nagaraja only focused on link additions toward nodes with high centrality values (e.g., degree centrality). Waniek *et al.* [26] proposed a heuristic algorithm by disconnecting internally and connecting externally (DICE) for hiding communities. Fionda and Pirro [1] proposed the concept of community deception to deal with community detection problem. Safeness-based deception algorithm and modularity-based deception algorithm were mainly tested through amounts of experiments.

Table I shows the comparisons between our approach and related studies. Our approach does not adopt the modularity method, but we improve Fionda's algorithm and design a new gain function expression to form a new community hiding algorithm called *Hs* that is based on a new safeness definition (i.e., safeness-based community hiding algorithm). For the new safeness definition, we give out the detailed theoretical proofs and formula demonstration in the following sections.

III. PRELIMINARIES AND PROBLEM FORMULATION

We first give preliminary definitions and related notations.

Definition 1: An undirected social network is denoted by $G = (V, E)$, where V is the node set and E is the link set. $|V|$ and $|E|$ represent the number of nodes and links in network G, respectively.

Definition 2: A set of communities discovered by a specific community detection algorithm A_D is denoted by $\bar{C} = \{C_1, C_2, \ldots, C_k\}$, where $C_i \subseteq V$, $C_1 \bigcup \ldots \bigcup C_k = V$ and $C_i \cap C_j = \emptyset$, for all $i, j \in \{1, \ldots, k\}$ and $i \neq j$. Given a community $\tilde{C} \subseteq V$, an $intra$-$\tilde{C}$ link (or intracommunity link) has the form $(u, v) : \{u, v\} \subseteq \tilde{C}$, while $inter$-$\tilde{C}$ link (or intercommunity link) has the form $(u, v) : u \in \tilde{C}, v \notin \tilde{C}$.

Definition 3: The degree of the dispersion of a community $\tilde{C}$ is denoted by $\rho(\tilde{C}) = \sum_u \sum_v sp(u, v)$, where $u, v \in \tilde{C}$ and $u \neq v$, and $sp(u, v)$ represents the length of the shortest path between nodes u and v. In our work, there must be a shortest path between any two nodes within community $\tilde{C}$. Because the *target community* we study must satisfy the basic condition of *reachability preservation*, the members of the *target community* belong to the same connected graph. On this condition, nodes within a *target community* can communicate to each other via the one or more link paths within community.

A. Problem Formulation

Community detection problem usually refers to detecting the closely connected parts of the network and these parts are called communities, that is, the connections within the communities are dense and the connections between communities are sparse. However, community hiding is the opposite of community detection and we must break the above nature of the community to resist community detection tools.

Although community detection is a problem that is easy to understand and study, scholars have done very little in terms of technology that allows the *target community* C to be hidden into multiple other communities. The community hiding problem can be described as: given a social network G and the community divisions $\bar{C}$ corresponding to a community detection algorithm A_D, we randomly select a community from $\bar{C}$ as the *target community* C and it is more in line with the practical significance of community hiding. The goal is to find a community hiding algorithm to hide the members of community C into other communities of the network G by perturbing a fraction of the social links, finally generating an updated network G'. After community hiding, algorithm A_D generates a new community structure on network G' and it is within this new community structure that the *target community* hiding is achieved.

This article proposes a community hiding algorithm that is based on link perturbation and designs a gain function to guide us to hide the *target community* into other communities by perturbing a budget (β) limited number of social links. Hence, the community hiding problem is equivalent to dealing the optimization gain function(ϕ) problem in (1) (Section IV gives the specific definition of the gain function), where $E(C)$ [resp., $\tilde{E}(C)$] represents the set of $intra$-C (resp., $inter$-C) links, E^+ (resp., E^-) represents a set of link additions (resp., deletions), and $|E^+| + |E^-| \leq \beta$. $E'(C)$ [resp., $\tilde{E}'(C)$] is the set of $intra$-C (resp., $inter$-C) links after link perturbation updates. The function ϕ formalizes the community hiding problem and the budget β controls the number of link updates. Hence, our work is just to pick the β changes that maximize ϕ

$$\underset{\{E'(C), \tilde{E}'(C)\}}{\arg\max} \quad \{\phi(C, E(C), \tilde{E}(C), \beta, E'(C), \tilde{E}'(C))\}$$

where

$$E'(C) \cup \tilde{E}'(C) = (E(C) \cup \tilde{E}(C) \cup E^+) \setminus E^-$$

and

$$|E^+| + |E^-| \leq \beta. \tag{1}$$

Fig. 1 shows the schematic of community hiding problem. Fig. 1(a) shows the input (a social network G) of community

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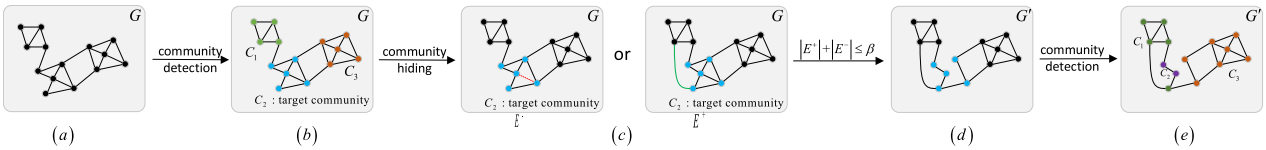

Fig. 1. (a) is the input (a social network G) of community hiding module. (b) is the community structure partition of G under a community detection algorithm and community C_2 is regarded as the target community. (c) is the step of the community hiding. The red dotted link represents a link deletion operation, and the green link represents a link addition operation. The deleted links are defined as E^- and the added links are defined as E^+. (d) is the output network (G') of target community hiding deception. (e) means that the community detection algorithm can't identify the community structure of the original target community members in updated network G'.

TABLE II
ADOPTED SYMBOLS

Symbols	Definition
$G(G')$	A (An updated) social network
A_D	A community detection algorithm
$C(C')$	The (updated) *target community*
$\bar{C}$	The community set
$\tilde{C}, C_i$	A community
u, v, w	The node of a network
n	The number of nodes in C
m	The number of links in C
$deg(u)$	The degree of node u
$\rho(\tilde{C})$	The degree of dispersion of $\tilde{C}$
$sp(u, v)$	The shortest path between nodes u and v
β	The link perturbation budget
$\psi(C)$	The value of intra_Community_Safeness for C
$\varphi(C)$	The strength of the connection between C and other nodes
$\sigma(C)$	The community safeness of C
$\Delta(C)$	The safeness gain
$\sigma(u, C)$	The node safeness
$M_G(\bar{C})$	The network partition modularity into $\bar{C}$
ML	The modularity loss
$R(C_i, C)$	The *recall* of A_D w.r.t C
$P(C_i, C)$	The precision of C_i in C
$H(C, \bar{C})$	The community Deception Score of C in $\bar{C}$

hiding module. Fig. 1(b) shows the community structure partition of G under a community detection algorithm and community C_2 is regarded as the *target community*. Fig. 1(c) shows the steps of the community hiding. The red dotted link represents a link deletion operation, and the green link represents a link addition operation. The delete links are defined as E^- and the added links are defined as E^+. Fig. 1(d) shows the output network (G') of *target community* hiding deception. Fig. 1(e) means that the community detection algorithm cannot identify the community structure of the original *target community* members in G'. Table II describes all the symbols used in this article.

IV. ANALYSIS AND COMMUNITY HIDING ALGORITHM

Given a *target community* C in a social network G, how can we hide the community in the whole network? We are committed to two operations: one is to make C dispersive by deleting the links in C and one is to increase the connection between the members of community C and the nodes outside community C. To illustrate the influence of the above two operations on community hiding, we analyze it from the aspects of $intra$-C and $inter$-C, respectively.

A. Concept of $intra$-C

First, we have to establish a consistent view: the best situation is that there are as few $intra$-C links as possible.

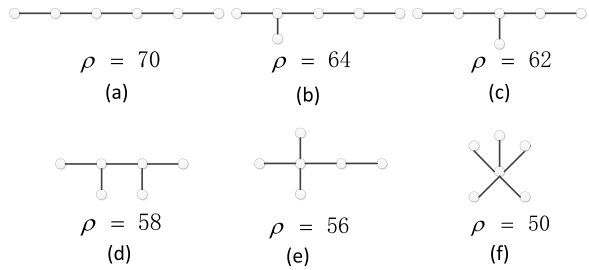

Fig. 2. When a network has 6 nodes and 5 links, (a) is a network structure where $\rho = 70$; (b) is a network structure where $\rho = 64$; (c) is a network structure where $\rho = 62$; (d) is a network structure where $\rho = 58$; (e) is a network structure where $\rho = 56$; (f) is a network structure where $\rho = 50$.

Thus, we can delete the links within C on the premise of not destroying the connected component. We believe that disrupting community connectivity will damage information interaction between community members and it may cause irreparable damage to C. Here, the number of nodes in C is denoted by $n = |C|$ and the number of links in C is denoted by m at first, so we have $m \geq n - 1$. In terms of $intra$-C, we are committed to delete the links in C to make the number of links in C close to $n - 1$, where the ideal situation is equal to $n - 1$. Its purpose is to make the contact within the *target community* as small as possible to fully resist the community detection tools. At this time, it meets the *reachability preservation* goal requirement. In our work, the degree of dispersion of a community C is defined as $\rho(C)$. In order to hide the community C, $\rho(C)$ must increase.

Taking Fig. 2 as an example, a *target community* C, only considering $intra$-C now, has $n = 6$ nodes and only one connected component. The number of links in C is greater than or equal to $n - 1 = 5$. We delete some links in C and assume that it is achievable that the number of links in C can be reduced to 5, and then, we can obtain one of the situations in Fig. 2 (due to the page requirements, only one graph is listed for each of different ρ). ρ of community in Fig. 2(a)–(f) is different. $\rho(C)$ of Fig. 2(a) equals 70 that is the maximum, whereas Fig. 2(f) equals 50 that is the minimum, that is, the degree of dispersion of Fig. 2(a) is the best, while Fig. 2(f) is the worst for community hiding. The former is conducive to community hiding and the latter is conducive to community mining. Hence, the only thing we need to do is to increase $\rho(C)$ of $intra$-C by link perturbation updates.

In this article, we normalize $\rho(C)$ into another indicator intra_Community_Safeness, which is defined as $\psi(C) = (\rho(C) - \rho(C)_{\min})/(\rho(C)_{\max} - \rho(C)_{\min})$, where $\rho(C)_{\min} = 2(n-1)^2$ [when one node is the center and the rest of the nodes are all connected to the center node, such as Fig. 2(f)] and $\rho(C)_{\max} = \sum_{k=1}^{n}(\sum_{i=1}^{n-k} i + \sum_{j=0}^{k-1} j)$ [when all nodes

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in the community are on a line, such as Fig. 2(a)]. The calculation formula of $\psi(C)$ is shown by (2). We assume that $\rho(C)$ has its maximum value when all nodes in C are located on a line as shown in Fig. 2(a); at this moment, $\psi(C) = 1$. Similarly, $\psi(C) = 0$ when $\rho(C) = \rho(C)_{\min}$. Here, we improve the intra_Community_Safeness of [1], which proved the infeasibility of $intra$-C link additions and $inter$-C link deletions

$$\psi(C) = \frac{\rho(C) - 2(n-1)^2}{\sum_{k=1}^{n}\left(\sum_{i=1}^{n-k} i + \sum_{j=0}^{k-1} j\right) - 2(n-1)^2}. \tag{2}$$

B. Concept of $inter$-C

Next, we analyze the connection between one member of a *target community* C and one external member ($inter$-C) of other communities. To hide the members of C into other communities as much as possible, it is necessary to increase contact with the outside nodes; namely, we can add links between nodes in C and nodes outside C. In this way, the members of C will be dispersed into other communities as much as possible and it meets *community spread* goal requirement. Then, how to evaluate the strength of the contact? Here, following the definition of [1], (3) represents the definition, where $u \in C$, and we denote $E(u, C)$ [resp., $\tilde{E}(u, C)$] as the set of intracommunity (*resp.*, intercommunity) links for node $u \in C$

$$\varphi(C) = \sum_{u \in C} \frac{|\tilde{E}(u, C)|}{\deg(u)}. \tag{3}$$

Equation (3) shows the ratio of u's links that point outside C to its degree and indicates an account on how u is hidden inside the network about its degree. In order to increase $\varphi(C)$, u should diversify its connections, that is, obtaining more connections with nodes outside of C by adding new links.

Under the premise of the *reachability preservation* and *community spread*, we can disperse all members of a *target community* into as many as other communities possible. In the end, we can achieve the purpose of *community hiding*.

C. Community Safeness

Taking $intra$-C and $inter$-C into account from the above discussion, we advocate the safeness of community consisting of both factors of $intra$-C and $inter$-C. Hence, we give a new definition referred to as *community safeness*, which is defined by the following equation:

$$\sigma(C) = \frac{1}{2}\psi(C) + \frac{1}{2}\varphi(C). \tag{4}$$

For community hiding problem, *Hs* designs a safeness gain $\Delta(C) = \sigma(C') - \sigma(C)$, where C' is the community after applying one or more link perturbation updates. Therefore, the greedy *Hs* algorithm selects the most appropriate link update that gives the highest $\Delta(C)$ at each step.

Now, we discuss the effect of link updates on community safeness and link additions and deletions separately.

1) Link Additions: First, we analyze the $inter$-C link additions. The *target community* C has community safeness $\sigma(C)$. Let $\Delta(C) = \sigma(C') - \sigma(C)$ be the safeness gain.

Theorem 1: For any $inter$-C link addition (u, w) $s.t.$ $u \in C$ and $w \notin C$ giving $G' = (V, E \cup (u, w))$, we have that: 1) $\Delta(C) \geq 0$ always holds and 2) the maximum increase in safeness happens for the node $u \in \arg\min\{((|\tilde{E}(u, C)|)/(\deg(u)))\}$.

Proof: Checking if $\Delta(C) = \sigma(C') - \sigma(C) \geq 0$ amounts at checking whether

$$\begin{aligned}\Delta(C) &= \sigma(C') - \sigma(C)\\ &= \frac{1}{2}(\psi(C') - \psi(C)) + \frac{1}{2}\frac{|\tilde{E}(u, C)|+1}{\deg(u)+1} - \frac{1}{2}\frac{|\tilde{E}(u, C)|}{\deg(u)}\\ &\geq 0.\end{aligned} \tag{5}$$

Since $\psi(C') = \psi(C)$ only considers $inter$-C link additions, $\Delta(C)$ can be written as

$$\begin{aligned}\Delta(C) &= \frac{1}{2}\frac{|\tilde{E}(u, C)| + 1}{\deg(u)+1} - \frac{1}{2}\frac{|\tilde{E}(u, C)|}{\deg(u)}\\ &= \frac{1}{2}\frac{\deg(u) - |\tilde{E}(u, C)|}{\deg(u)(\deg(u)+1)}.\end{aligned} \tag{6}$$

Condition 1) (i.e., $\Delta(C) \geq 0$) holds since $\deg(u) \geq |\tilde{E}(u, C)|$. Moreover, condition 2) means that the maximum increase happens. □

Theorem 2: An $intra$-C link addition (u, w) s.t. $\{u, w\} \subset C$ giving an updated network $G' = (V, E \cup \{(u, w)\})$ always gives $\Delta(C) < 0$.

Proof: There is a truth that $\psi(C') \leq \psi(C)$ when adding an $intra$-C link, so we note that the following condition always holds:

$$\begin{aligned}\Delta(C) = \sigma(C') - \sigma(C) &= \frac{1}{2}(\psi(C') - \psi(C))\\ &+ \frac{1}{2}\left(\frac{|\tilde{E}(u, C)|}{\deg(u)+1} - \frac{|\tilde{E}(u, C)|}{\deg(u)}\right)\\ &+ \frac{1}{2}\left(\frac{|\tilde{E}(w, C)|}{\deg(w)+1} - \frac{|\tilde{E}(w, C)|}{\deg(w)}\right) < 0.\end{aligned} \tag{7}$$

□

2) Link Deletions: We now pay attention to link deletions and give our theories and proofs.

Theorem 3: An $inter$-C link deletion $(u, w) : u \in C, w \notin C$ giving $G' = (V, E\backslash\{(u, w)\})$ always gives $\Delta(C) \leq 0$.

Proof: To prove $\Delta(C) = \sigma(C') - \sigma(C) \geq 0$ amounts at checking whether

$$\frac{1}{2}(\psi(C') - \psi(C)) + \frac{1}{2}\frac{|\tilde{E}(u, C)| - 1}{\deg(u) - 1} - \frac{1}{2}\frac{|\tilde{E}(u, C)|}{\deg(u)} \geq 0. \tag{8}$$

Similar to Theorem 1, an $inter$-C link deletion has no effect on the shortest path between the inner node pairs of C, that is, $\psi(C') = \psi(C)$. Therefore, (8) clearly never holds since $\deg(u) \geq |\tilde{E}(u, C)|$. □

Theorem 4: An $intra$-C link deletion $(u, w) : \{u, w\} \subseteq C$ always gives $\Delta(C) > 0$.

Proof: First of all, we must set a premise. The nodes u and w in C still belong to the same connected component

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after the deletion of the link (u, w) because we cannot destroy the connectivity of the *target community*. Then, similar to Theorem 2, $\psi(C') \geq \psi(C)$ when deleting an $intra$-C link. Therefore, we prove that the following condition always holds:

$$\begin{aligned}\Delta(C) = \sigma(C') - \sigma(C) &= \frac{1}{2}(\psi(C') - \psi(C)) \\ &+ \frac{1}{2}\left(\frac{|\tilde{E}(u, C)|}{\deg(u) - 1} - \frac{|\tilde{E}(u, C)|}{\deg(u)}\right) \\ &+ \frac{1}{2}\left(\frac{|\tilde{E}(w, C)|}{\deg(w) - 1} - \frac{|\tilde{E}(w, C)|}{\deg(w)}\right) > 0. \end{aligned} \tag{9}$$

□

Algorithm 1 *Hs*—Community Hiding via Safeness

Input: $G = (V, E), C, \beta$
Output: G'
1: **do**
2: $n_p = getNodeMinimumAddRatio(C)$ /*Theorem 1*/
3: $n_t = findRandomExternalNode(n_p, C, G)$
4: $\Delta(C)^{add} \leftarrow getAdditionGain(n_p, n_t, C)$
5: $(n_k, n_l) = getBestDelExclBridges(C)$ /*Theorem 4*/
6: $\Delta(C)^{del} \leftarrow getDeletionGain(n_k, n_l, C)$
7: **if** $\Delta(C)^{add} \geq \Delta(C)^{del}$ and $\Delta(C)^{add} > 0$ **then**
8: $G \leftarrow (V, E \cup \{(n_p, n_t)\})$
9: **else if** $\Delta(C)^{del} > 0$ **then**
10: $G \leftarrow (V, E \setminus \{(n_k, n_l)\})$
11: $\beta = \beta - 1, G' = G$
12: **while** $\beta > 0$ and $(\Delta(C)^{add} > 0$ or $\Delta(C)^{del} > 0)$

In summary, the community hiding algorithm *Hs* is outlined in Algorithm 1 according to the above analysis [1]. In the experiment, our goal is to increase $\sigma(C)$. The algorithm only considers $inter$-C link additions in terms of additions according to Theorem 1 because it always brings a safeness gain. According to Theorem 2, Algorithm 1 does not consider the $intra$-C link additions because $\Delta(C)$ is always less than 0. Similarly, for link deletions, under the premise of keeping the community connected, the algorithm only considers $intra$-C link deletions according to Theorem 4 and discards the $inter$-C link deletions according to Theorem 3. Besides, this algorithm only requires $intra$-C and $inter$-C links.

The function $getNodeMinimumAddRatio$ (line 2) computes, for every single node $u \in C$, the ratio of u's links that point outside C to its degree and then select the node n_p corresponding to the minimum fraction. The function $findRandomExternalNode$ (line 3) randomly finds a node n_t (not in community C) such that the link (n_p, n_t) does not exist. The function $getAdditionGain$ (line 4) computes the link addition gain according to lines 2 and 3. The function $getBestDelExclBridges$ (line 5) consists of two steps: one step is that it excludes the social links that could destroy the connectivity of C on condition that they are deleted and the other step is that it calculates $\Delta(C)$ according to Theorem 4 for each remaining link. The function $getDeletionGain$ (line 6) computes the link deletion gain according to line 5. Finally, *Hs* chooses the most suitable link update and conducts the update on network (lines 7–11). For each loop operation, we can obtain the best node for link addition and the best link for deletion by checking all the nodes and links in C, so the time complexity of Algorithm 1 is $O(\beta(n + m))$.

V. Experimental and Experimental Evaluation

A. Baselines

To illustrate the effectiveness and feasibility of our method, in this article, we introduce two algorithms as baselines: *Ds* (Safeness-based community deception) and *Dm* (Modularity-based community deception) algorithms, and both of them are defined in [1].

Baseline 1 (Ds): Fionda and Pirro [1] proposed the *Ds* algorithm. The difference between *Ds* and *Hs* is that *Hs* considers the total shortest path for each pair of nodes in *target community* C, whereas *Ds* considers whether one node in C can reach another node in C only with the help of nodes in C. First, *Ds* introduces the concept of node safeness $\sigma(u, C)$. The node safeness $\sigma(u, C)$ of u in G is shown as follows:

$$\sigma(u, C) = \frac{1}{2}\frac{\left|V_C^u\right| - |E(u, C)|}{|C| - 1} + \frac{1}{2}\frac{|\tilde{E}(u, C)|}{\deg(u)} \tag{10}$$

where $V_C^u \subseteq C$ is the set of nodes reachable from u passing only via nodes in C.

Furthermore, the community safeness $\sigma(C)$ is defined as follows:

$$\sigma(C) = \sum_{u \in C} \sigma(u, C)/|C|. \tag{11}$$

For the general deception formulation, *Ds* proposes a safeness gain $\xi_C = \sigma(C') - \sigma(C)$ and it chooses the link update that gives the highest ξ_C at each step through the greedy algorithm (community safeness). The budget is also defined as β. *Ds* algorithm runs in time $O(\beta(n + m))$.

According to (11), Fionda and Pirro [1] proved the effect of link updates (link additions and link deletions) on community safeness; $inter$-C link addition has a positive influence on community hiding, whereas $inter$-C link deletion has a negative effect on community hiding. However, the authors points out that $intra$-C link addition and $intra$-C link deletion do not always introduce a safeness gain. Both of these updates have a negative impact on the community hiding under certain circumstances. In the experiment, the authors do not consider the $intra$-C link addition because they assume that it is a decision of C's members to keep the community disconnected if C is disconnected from the beginning. Besides, Fionda and Pirro [1] considered the $intra$-C link deletion under certain conditions. Obviously, different from [1], our method can prove exactly which updates have a positive or negative impact on community hiding so that there is no need to add artificial assumptions.

Baseline 2 (Dm): Except the *Ds* algorithm, Fionda and Pirro [1] also gave the modularity-based deception algorithm *Dm*. It is denoted by $\deg(v)$ the degree of $v \in V$. The degree of a community C_i is denoted by: $\deg(C_i) = \sum_{v \in C_i} \deg(v)$. The modularity of the partition of a network into community $\bar{C} = \{C_1, C_2, \ldots, C_k\}$ is given by the following equation:

$$M_G(\bar{C}) = \frac{\eta}{m} - \frac{\delta}{4m^2} \tag{12}$$

where $\eta = \sum_{C_i \in \bar{C}} |E(C_i)|$ and $\delta = \sum_{C_i \in \bar{C}} \deg(C_i)^2$.

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TABLE III
USED SOCIAL NETWORKS IN THIS WORK

Net	$\|V\|$	$\|E\|$	Description
kar	34	78	Zachary karate's club [34], [35]
dol	62	159	Dolphins association [36]
footb	115	613	American football games [35]
polb	105	441	Books about US politics
net	1589	2742	The coauthorship network [33]
faceb	4039	88234	The Facebook dataset [34], [35]
pow	4941	6594	USA power grid [37]
erd	6100	9939	Erdos collaborations network
ast	17.9k	197k	Astrophysics coauthorship network [38]

For community hiding, an intuitive strategy would be to leverage the modularity loss $\mathrm{ML} = M_G(\bar{C}) - M_{G'}(\bar{C})$, where G' is the network after applying link perturbation. The goal of *Dm* is to find the set of β link updates where ML is maximized. *Dm* algorithm runs in time $O(|E| + |\bar{C}|^2)$.

B. Community Detection Algorithms

In the field of community detection algorithms, in this article, we consider five mature and available detection algorithms: Label Propagation algorithm [27], InfoMap algorithm [28], [29], MultiLevel algorithm [30], SpinGlass algorithm [31], [32], and Leading Eigenvectors algorithm [33]. They are all available in the Python version of the igraph library.[1]

C. Data Sets

The data sets used in this work are many real social networks, which are free for scientific use and can be downloaded online[2,3,4] and the original authors have already made the data sets freely available. It includes both small-scale (kar, dol, $footb$, and $polb$) and large-scale (net, $faceb$, pow, erd, and ast) social networks. Table III gives a description of the all used social networks for experiments.

D. Evaluation Metric

Here, we use the *deception score* H to describe the quality of our algorithm. Fionda and Pirro [1] defined the H score in detail. For a certain community detection algorithm, we can get a community structure $\bar{C} = \{C_1, C_2, \ldots C_k\}$ of a certain network G. First, the $recall$ of a detection algorithm A_D with respect to a *target community* C is defined as follows:

$$R(C_i, C) = \frac{\#C's \text{ members in } C_i \text{ found by } A_D}{|C|} \quad \forall C_i \in \bar{C}. \tag{13}$$

Then, the $precision$ is defined as follows:

$$P(C_i, C) = \frac{\#C's \text{ members in } C_i \text{ found by } A_D}{|C_i|} \quad \forall C_i \cap C \neq \emptyset \tag{14}$$

[1]https://igraph.org/python/doc/igraph.Graph-class.html
[2]http://www-personal.umich.edu/~mejn/netdata/
[3]http://snap.stanford.edu/data/index.html
[4]http://www.orgnet.com/

According to (13) and (14), the community *deception score* H is defined as

$$H(C, \bar{C}) = \left(1 - \frac{|S(C) - 1|}{|C| - 1}\right) \times \left(\frac{1}{2}(1 - \max_{C_i \in \bar{C}} \{R(C_i, C)\}) + \frac{1}{2}\left(1 - \frac{\sum_{C_i \cap C \neq \emptyset} P(C_i, C)}{|C_i \cap C \neq \emptyset|}\right)\right) \tag{15}$$

where $|S(C)|$ is the number of connected components in the subgraph induced by $C's$ members. H ranges from 0 to 1. $H = 1$ represents that the nodes in C are in a single connected component and are hidden in the network in the best way. $H = 0$ represents that each node in C comes from different components or $C \in \bar{C}$.

E. Evaluation Algorithm

According to H, Fionda and Pirro [1] introduced a standard benchmark algorithm to test deception algorithms and we follow this evaluation algorithm. Similar to [1], we apply a budget of changes β such that $|E^+| + |E^-| \leq \beta$ to the network G, and then, we obtain an update network G' after executing the community hiding algorithms (*Hs*, *Ds*, and *Dm*).

Algorithm 2 Evaluating Community Hiding Algorithm

1: **procedure** EVALUATEHIDINGALGOG(β,A_D)
2: $\bar{C} \leftarrow A_D(G)$ /*Communities before hiding*/
3: $C \leftarrow gettargetCommunity(\bar{C})$
4: $M_G(\bar{C}) \leftarrow getModularity(\bar{C}, C)$
5: $\sigma(C) \leftarrow getSafeness(C, G)$
6: $H(C, \bar{C}) \leftarrow getHiding(C, G, \bar{C})$
7: $G' \leftarrow applyHiding(G, C, \beta, Hs|Ds|Dm)$
8: $\bar{C}' \leftarrow A_D(G')$ /*Communities after hiding*/
9: $M_{G'}(\bar{C}') \leftarrow getModularity(\bar{C}', G')$
10: $\sigma(C') \leftarrow getSafeness(C, G')$
11: $H(C, \bar{C}') \leftarrow getHiding(C, \bar{C}', G')$

Algorithm 2 starts with a community division $\bar{C}$ found by given community detection algorithm A_D (line 1). In order to better show the hidden effect, we select a fully discovered community in $\bar{C}$ as the *target community* C (i.e., $C \in \bar{C}$) (line 2). Lines 4–6 compute the original values of modularity, safeness, and H score. Then, we apply community hiding algorithms (*Hs*, *Ds*, and *Dm*), respectively, on *target community* and obtain the updated network G' (line 6). In this way, a new community structure $\bar{C}'$ can be obtained for network G' by algorithm A_D (line 7) and compute the values of modularity, safeness, and H score again to measure the impact of the hiding strategies (lines 8 and 9).

F. Results Analysis

We run all community hiding methods on the given data sets and the experiments are run on a server of 3.50-GHz CPU and 16.00-GB RAM.

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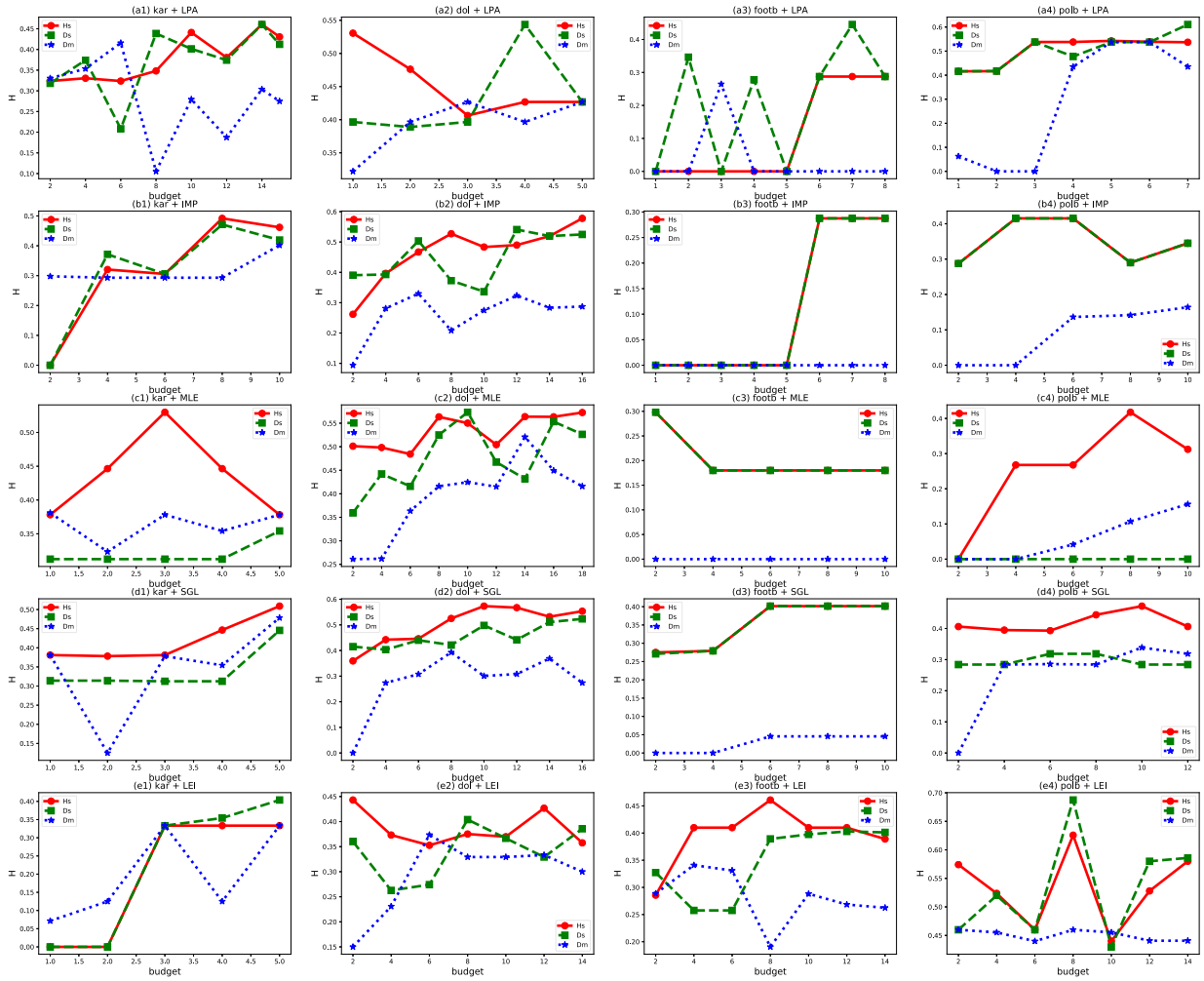

Fig. 3. Evolution of deception score H as a function of the budget on many small-scale networks.

Fig. 3 shows a detailed view of the deception score when considering *Hs*, *Ds*, and *Dm*, and budget β depends on the size of the selected *target community* C. If $n = |C|$, then β are $1, 2, 3, \ldots,$ when $n < 10$ and β are $2, 4, 6, \ldots, n$ when $n \geq 10$.

First, Fig. 3 contains a total of 20 (five rows and four columns) detailed result comparison subfigures for small-scale social networks, and each subfigure in Fig. 3 shows the evolution of H scores under the three community hiding algorithms (i.e., *Hs*, *Ds*, and *Dm*) with different budgets. As shown, the results of subfigures in each row from top to bottom represents the impact of the five community detection algorithms (*LPA*, *IMP*, *MLE*, *SGL*, and *LEI*) on the H score of the three community hiding algorithms, and each column from left to right presents results for different social networks. On the whole, the *Hs* and *Ds* algorithms perform better than *Dm* algorithm, and the *Hs* algorithm outperforms *Ds* algorithm.

In detail, for small-scale social networks, *Hs* and *Ds* are better than *Dm*. In Fig. 3, we note that the H score polyline of *Dm* algorithm below the H score polyline of *Hs* and *Ds* algorithms in over 90% of cases. Also, we observe a bad situation that *Dm* algorithm may be useless for community hiding ($H = 0$) once in a while. The reason for such phenomenon is mainly due to these detection algorithms that are not based on modularity optimization algorithm, which generates that an updated network does not bring enough effect on the function because of the low modularity. For *Hs* and *Ds* algorithms, we can observe that the *Hs* algorithm performs better than the *Ds* algorithm in most cases although there is a special counterexample [as shown in Fig. 3(a3)] that *Ds* is a bit better than *Hs* in case "$footb + LPA$." We can also see that in some cases, the H score of *Hs* and *Ds* is very near, such as these cases: "$footb + IMP$," "$polb + IMP$," "$footb + MLE$," and "$footb + SGL$"' in Fig. (b3), (b4), (c3), and (d3), respectively. The rest of the results show that *Hs* is significantly better than *Ds*.

Similarly, Fig. 4 clearly describes the community *deception score* H of *Hs*, *Ds*, and *Dm* algorithms for large-scale social networks (net, $faceb$, pow, erd, and ast) under five community detection algorithms (*LPA*, *IMP*, *MLE*, *SGL*, and *LEI*) (to be consistent with [1], *MLE* is not applied on data sets pow, erd, and ast, and *SGL* is not applied on net and $faceb$). Intuitively, in terms of community hiding algorithms for

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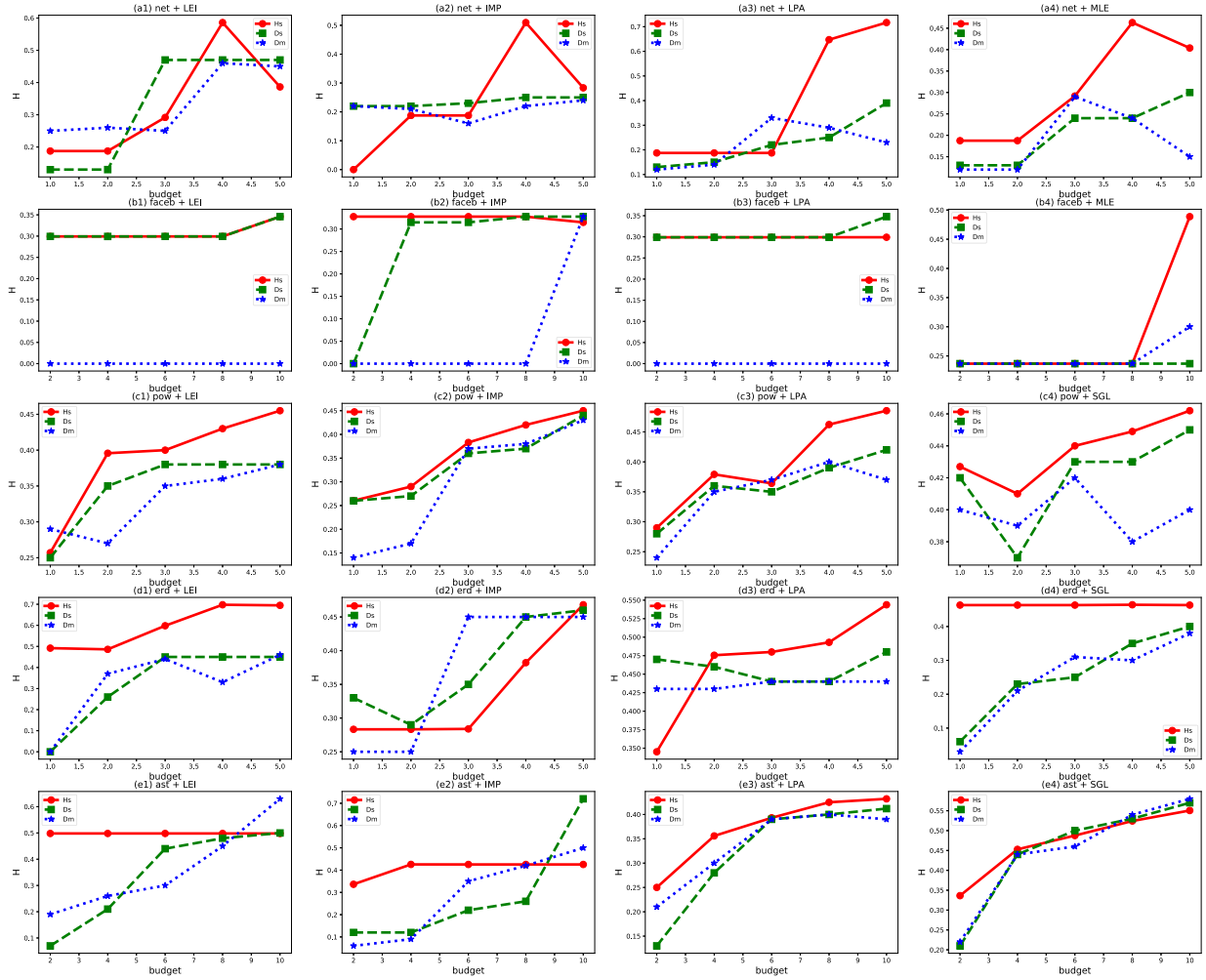

Fig. 4. Evolution of deception score H as a function of the budget on many large-scale networks.

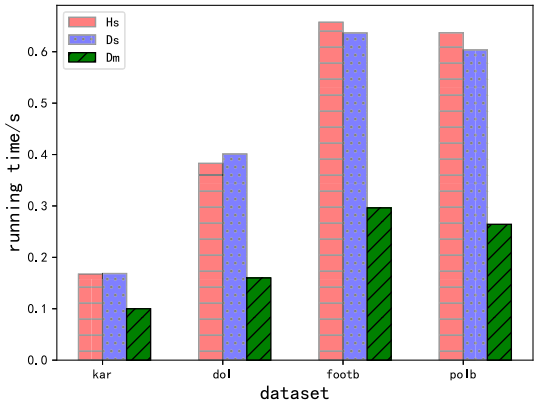

Fig. 5. Running time (in seconds) of *Hs*, *Ds*, and *Dm* on small-scale social networks.

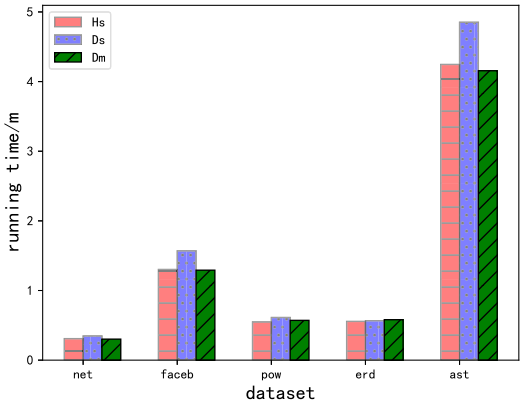

Fig. 6. Running time (in minutes) of *Hs*, *Ds*, and *Dm* on large-scale social networks by using log scale.

large-scale social networks, *Hs* performs better than *Ds* algorithm, and *Ds* performs better than *Dm* algorithm. First, for $faceb$ and pow social networks, *Hs* algorithm is significantly better than *Ds* algorithm except for a few special conditions, such as "$faceb + IMP$" (except when budget $= 10$) and "$faceb + LPA$" (except when budget $= 10$). Similarly, *Hs* is also significantly better than *Ds* and *Dm* for cases "$erd + LEI$" and "$erd + SGL$" in Fig. 4(d1) and (d4). Because the community hiding method relies on the original community divisions and the network structure, our Hs shows a bit performance reduction compared to the *Ds* and *Dm* algorithms such as case "$erd + IMP$" in Fig. 4(d2).

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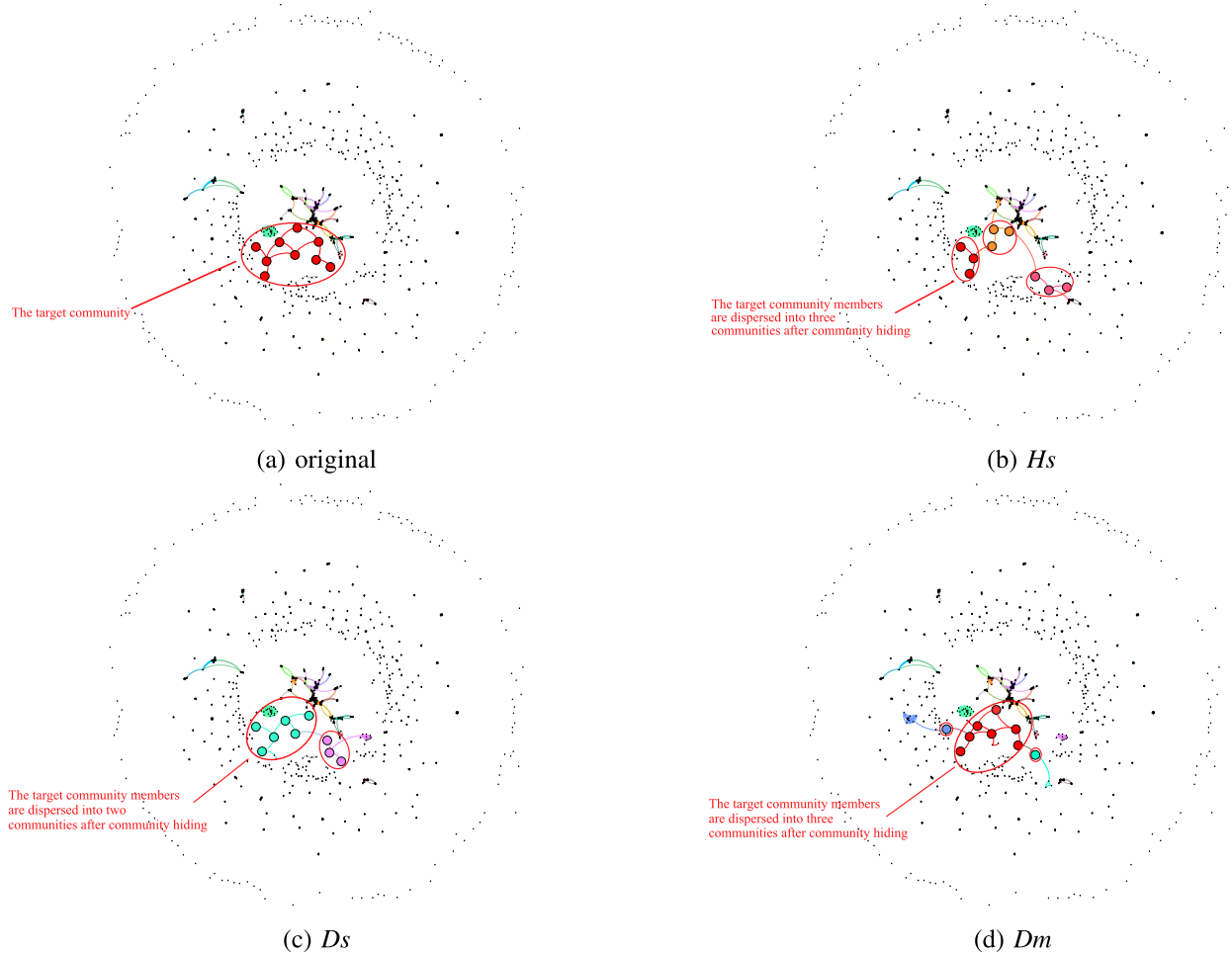

Fig. 7. Example to illustrate the difference among the three algorithms (*Hs*, *Ds*, and *Dm*). Graph (a) shows the original community partition result of the *net* network. (b), (c), and (d) Effect of the three community hiding algorithms on the *target community* C.

On the whole, our *Hs* method outperforms the two baselines in the above extensive experiments.

G. Running Time Evaluation

In this section, we evaluate the average running time of the three community hiding algorithms (*Hs*, *Ds*, and *Dm*) on the four small-scale social networks (kar, dol, $footb$, and $polb$) and the five large-scale social networks (net, $faceb$, pow, erd, and ast), and the runtime includes the time spent to load the social networks in memory and the time executing the community hidden operation. Figs. 5 and 6 show the comparison of the running time of the three community hiding algorithms on small-scale networks and large-scale networks, respectively.

Fig. 5 shows a detailed result of the running time, and we can find that all the experiments can be finished within one second. In general, the *Hs* and *Ds* algorithms require almost the same running time, whereas *Dm* algorithm is slightly better than *Hs* and *Ds*. However, since the H score of the *Dm* algorithm is generally too low, the performance of *Dm* is greatly reduced. With comprehensive performance and acceptable algorithm running time, *Hs* and *Ds* algorithms are better than *Dm* algorithm.

As shown in Fig. 6 (Fig. 6 uses a log scale on y-axis), the running time required for community hiding algorithms increases with the network size. When the network size is between 1000 and 10 000, the efficiency of the three community hiding algorithms is comparable. As a special case, the running time for network ast is much longer than the other four networks because its number of nodes and links is dozens of times of the other four networks.

H. Case Study

In this section, we give an example to illustrate the difference between the three community hiding/deception methods. Taking net network for an example, we select a *target community* $C \in \bar{C}$ corresponding to *LPA* detection algorithm, and the budget is 4. Fig. 7 intuitively shows the hiding effect of three community hiding algorithms on the *target community* in social networks and the members are marked in circles.

Fig. 7(a) shows the original division of the net social network. There are 451 communities in the original graph. The nodes with the same color come from the same community and the node set indicated by the red color is the *target community* C. Fig. 7(b)–(d) represents the community division results of the community hiding algorithm *Hs*, *Ds*, and *Dm*,

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respectively. Fig. 7 shows that the *Hs* algorithm distributes the members of the *target community* to other communities as much as possible and it meets the goal of *community spread* under the condition of *reachability preservation*. The effect of *Ds* algorithm is worse than that of *Hs* algorithm, and the *target community* members are only dispersed into two communities after community hiding. For *Dm* algorithm, the *target community* members are dispersed into three communities after community hiding. According to Fig. 7(d), *Dm* has no link perturbation operation within a community when hiding its members and this is also one of its disadvantages. Besides, as for *community spread*, *Hs* is better than *Dm* because most of the members of the *target community* are still in the same community in Fig. 7(d). In summary, compared with *Ds* and *Dm* algorithms, our *Hs* algorithm can hide the *target community* better.

VI. Conclusion

In this article, we have studied the community security problem and proposed a new improved algorithm (i.e., *Hs*) to hide the members of a sensitive *target community* into other nonsensitive communities of a social network. First, we converted the community hiding problem into an optimization problem by designing a concrete instantiation of gain function. Besides, to evaluate the community hiding algorithm, we followed the H score evaluation method that is proposed in [1]. Finally, we conducted extensive comparative experiments on a set of small- and large-scale social networks to demonstrate the effectiveness of our method. The results showed that our *Hs* method can achieve the highest H score when compared with two baselines (*Ds* and *Dm*) in most of the social community divisions. For running time, although *Hs* and *Ds* had similar running time consumptions on the small-scale networks, the *Hs* achieved higher efficiency than *Ds* on the large-scale social networks.

Our research only focuses on single *target community* deception problem, but privacy protection of social network structure contains three research ideas: 1) single *target community* deception problem; 2) multiple *target communities* deception problem; and 3) complete network structure deception problem. The first problem is the basis for studying other problems and the research of this article is the foundation of the privacy protection of social network structure. Ideas 2) and 3) are very challenging; it requires more energy and more comprehensive research in future.

Although *target community* deception can help users hide their community privacy, it will cause a bad effect if it is employed by the attackers or criminals. Therefore, we put forward the concept of community inference, which is different from community detection. Community inference needs to supplement missing information and identify false links by using node attributes information. For example, in a user relationship network, user interests, user location, and user occupation can be considered as the attributes because the members of a certain community must have very similar characteristics in some respects. The differences in these characteristics can help us to identify the false members in a community and to resist the hidden attacker community. Thus, community inference is a valuable research problem in future.

In summary, community hiding is of great significance in social security issues, and more related research will emerge out in many aspects: 1) designing new and more efficient solutions for the community hiding problem; 2) studying multiple *target community* deception problem; 3) inspiring novel and antihiding community detections to discover the hidden *target community* by using node attributes; and 4) real applications into the complex social system to protect individual and community privacy.

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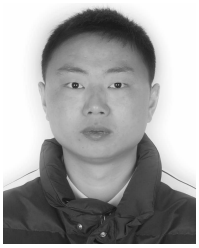

Xianyu Chen received the B.S. degree from Anhui University, Hefei, China, in 2018. He is currently pursuing the M.S. degree with the School of Cyber Engineering, Xidian University, Xi'an, China.

His current research interests include social network analysis and privacy protection.

Zhongyuan Jiang received both B.S. and Ph.D. degrees from Beijing Jiaotong University, Beijing, China, in 2009 and 2013, respectively.

Currently, he is an Associate Professor with the School of Cyber Engineering, Xidian University, Xi'an, China. His research interests include privacy preserving, social computing, urban computing, and network functions virtualization.

Hui Li received the B.S. degree from the Harbin Institute of Technology, Harbin, China, in 2005, and the Ph.D. degree from Nanyang Technological University, Singapore, in 2012.

He is a Professor with the School of Cyber Engineering, Xidian University, Xi'an, China, and a member of The State Key Laboratory of Integrated Services Network. His research interests include social computing, knowledge discovery, graph mining, and data privacy in big data.

Jianfeng Ma received the B.S. and Ph.D. degrees from Shaanxi Normal University, Xi'an, China, and Xidian University, Xi'an, China, in 1985 and 1995, respectively.

He is currently a Professor with the School of Cyber Engineering, Xidian University, and the Director of the Key Laboratory of Network and System Security of Shaanxi Province. His research interests include cryptography, wireless network security, data security, and mobile intelligent system security.

Philip S. Yu (Fellow, IEEE) received the Ph.D. degree from Stanford University, Stanford, CA, USA.

He is a Distinguished Professor and the Wexler Chair in Information Technology with the Department of Computer Science, University of Illinois at Chicago (UIC), Chicago, IL, USA. Before joining UIC, he was at the IBM Watson Research Center, Ossining, NY, USA, where he built a world-renowned data mining and database department. His research interests include data mining, social computing, and privacy protection.

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